UBICOMP ISWC2025

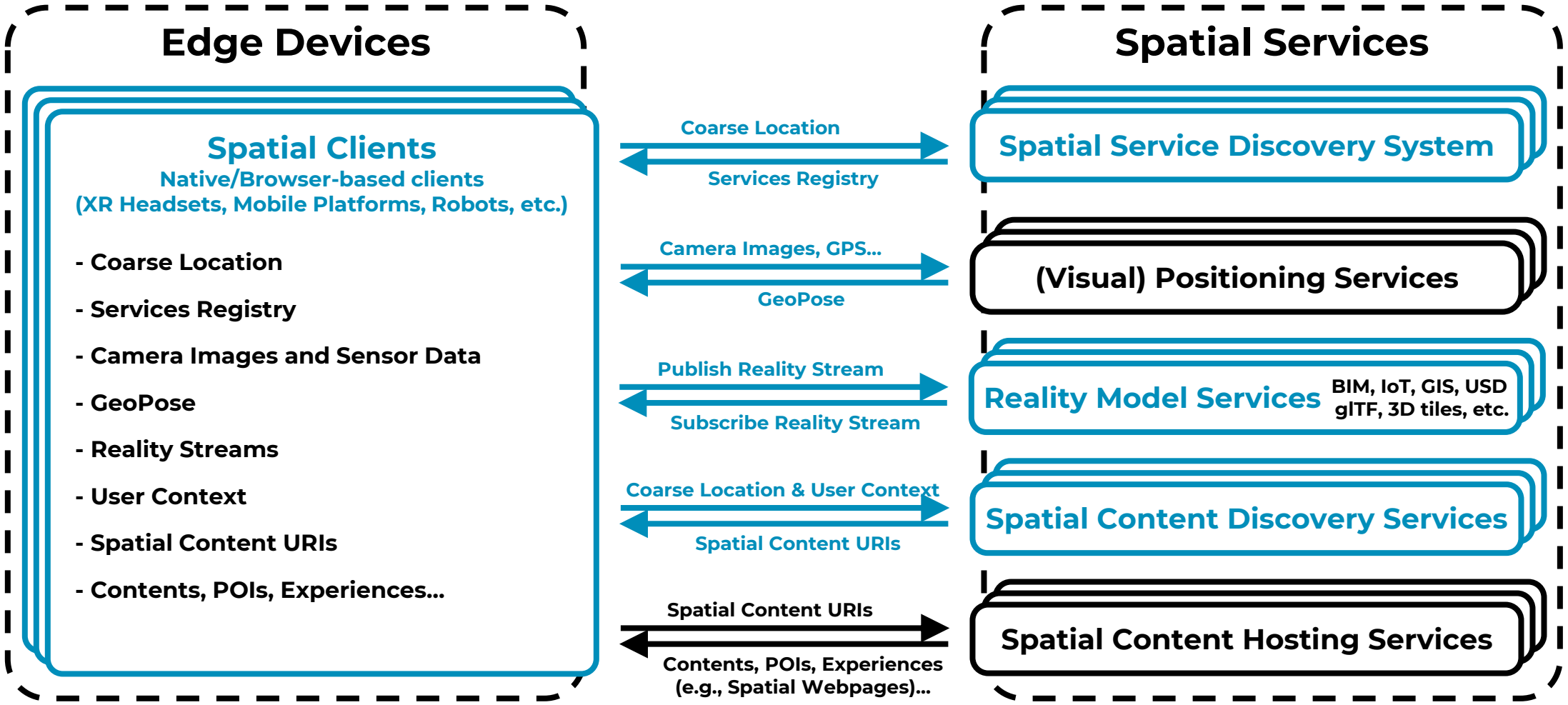
14-16 Oct. 2025
Helsinki, Finland

# An open spatial computing platform

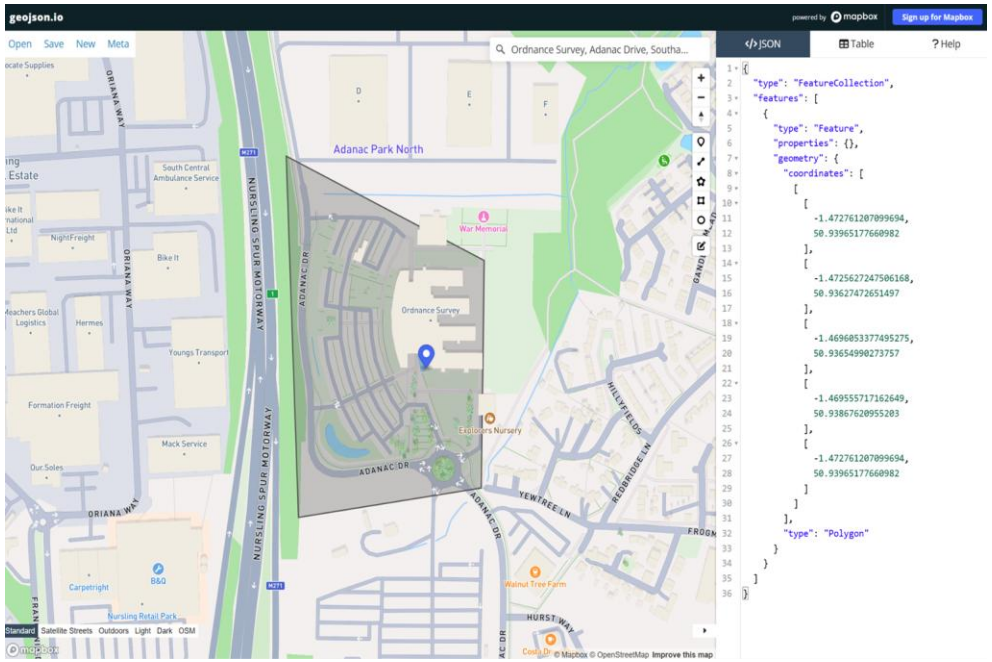
James Jackson

OPEN AR CLOUD

# OSCP Arch


Edge Devices
Spatial Clients
Native/Browser-based clients
(XR Headsets, Mobile Platforms, Robots, etc.)
- Coarse Location
- Services Registry
- Camera Images and Sensor Data
- GeoPose
- Reality Streams
- User Context
- Spatial Content URIs
- Contents, POIs, Experiences...
Coarse Location
Services Registry
Camera Images, GPS...
GeoPose
Publish Reality Stream
Subscribe Reality Stream
Coarse Location & User Context
Spatial Content URIs
Spatial Content URIs
Contents, POIs, Experiences
(e.g., Spatial Webpages)...
Spatial Services
Spatial Service Discovery System
(Visual) Positioning Services
Reality Model Services
BIM, IoT, GIS, USD
glTF, 3D tiles, etc.
Spatial Content Discovery Services
Spatial Content Hosting Services

# Spatial Discovery Demo




OSCP Spatial Discovery Gen 1 - HTTP, P2P, OSM

https://github.com/OpenArCloud/oscp-spatial-service-discovery
https://github.com/OpenArCloud/oscp-spatial-content-discovery

OSCP Spatial Discovery Gen 2 - HTTP, no P2P, Tile38

https://github.com/OpenArCloud/SSD-FastAPI-Tile38
https://github.com/OpenArCloud/SCD-FastAPI-Tile38

**Spatial Content Record (SCR)**

```
{
   "contenttype": "model/gltf+json",
   "description": "test model",
   "geopose": {
      "position": {
         "h": 1000.0,
         "lat": 50.93965177660982,
         "lon": -1.472761207099694
      },
      "quaternion": {
         "w": 0.0,
         "x": 0.0,
         "y": 0.0,
         "z": 0.0
      }
   },
   "metatype": "model3D",
   "tags": {
      "name1": "value1",
      "name2": "value2"
   },
   "url": "http://path_to_model"
  }
 ],
 "id": "uuid1"
}
```

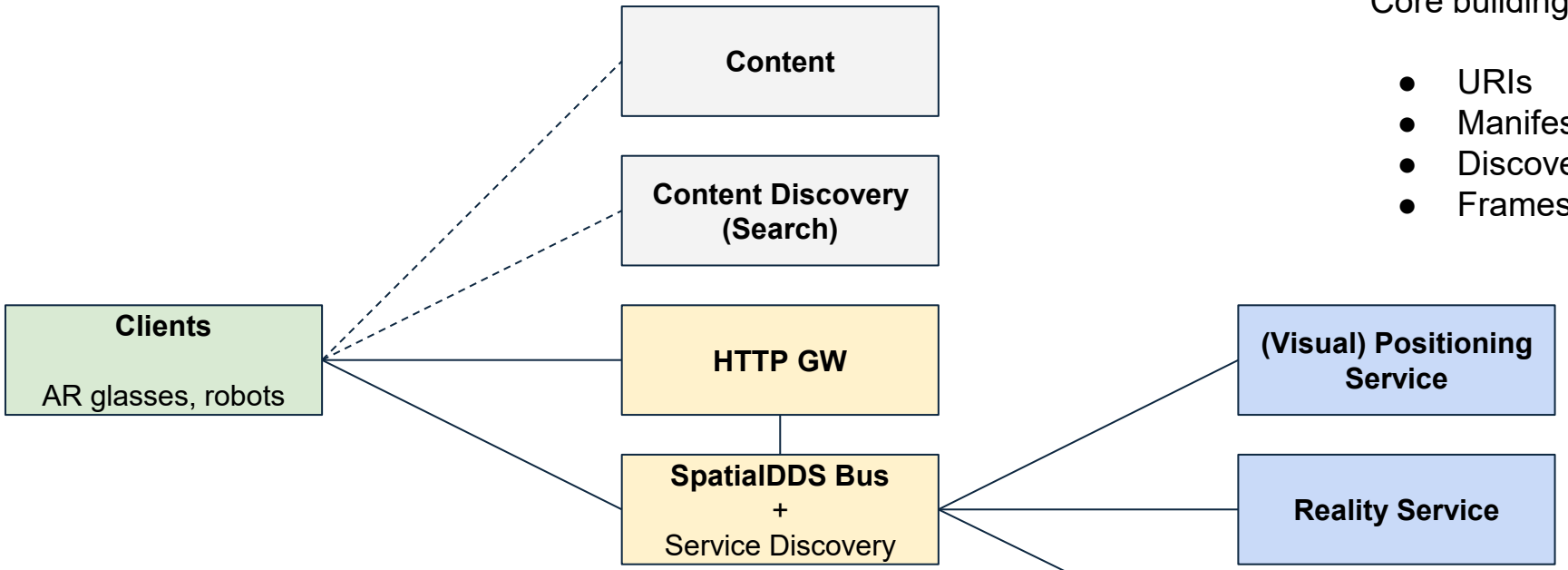
**Spatial Service Record (SSR)**

```
"properties": {
   "id": "uuid1",
   "services": [
      {
         "tags": {
            "name1": "value1",
            "name2": "value2"
         },
         "type": "geopose",
         "url": "http://www.geopose.org"
      }
   ]
}
```

# SpatialDDS (Data Distribution Service) - A Preview

Core building blocks:

- URIs
- Manifests
- Discovery
- Frames/Transforms




**Services**
(also communicate among themselves via SpatialDDS)

Domains:

- Robotics & Autonomous Vehicles
- Augmented & Mixed Reality
- Digital Twins & Smart Cities
- IoT & Edge AI
- AI World Models & Agents

# SpatialDDS Resources + Demo

Spec website: spatialdds.org
Spec repo: github.com/OpenArCloud/SpatialDDS-spec
Demo repo: github.com/OpenArCloud/SpatialDDS-demo

Contributors welcome!